

A spruce beetle susceptibility map of Alaska using Sentinel-1 radar and crown-scale mortality data

Journal:	*Forestry: An International Journal of Forest Research*
Manuscript ID	Forestry-2026-310
Manuscript Type:	Original Article
Date Submitted by the Author:	19-Oct-2026
Complete List of Authors:	Murphy, Seamus; TÜV SÜD, Forest Carbon Verification
Keywords:	Spruce beetle, Dendroctonus rufipennis, stand susceptibility, outbreak hazard rating, Sentinel-1, canopy height



A spruce beetle susceptibility map of Alaska using Sentinel-1 radar and crown-scale mortality data

Correspondence: Seamus Murphy, TÜV SÜD, 2187 Comox Ave, Comox, British Columbia V9M 1P5, Canada. seamusrobertmurphy@gmail.com. ORCID 0000-0002-1792-0351.

Abstract

Outbreaks of the spruce beetle (*Dendroctonus rufipennis* (Kirby)) in Alaska have killed the largest spruce first, so the tallest stands are most at risk in the next outbreak. Managers rate that hazard from tree diameter, basal area, and spruce share measured on the ground, which no cruise covers statewide, and no study had tested whether radar measures of stand structure pick out the stands an outbreak kills. This study scored Sentinel-1 radar backscatter of 2014 to 2016, before the current outbreak reached the Susitna valley, and a canopy height map against a crown-scale map of the spruce killed there by 2021 (41 764 cells of 90 m), then pooled every scene over Alaska into a 90 m map of pre-outbreak backscatter tested against the aerial detection survey. The killed stands had returned more backscatter before the outbreak and were taller, 24.7 m against 12.9 m. At the threshold that best separated the two, the backscatter map classed 91 per cent of the killed and 76 per cent of the spared stands correctly, an overall accuracy of 89 per cent. On the Kenai Peninsula and in the Matanuska-Susitna Borough it marked 77 and 80 per cent of the attacked forest as susceptible and 44 and 48 per cent of the unattacked, respectively. In the Copper River basin, where attacked stands were 13.6 m tall against 10.4 m, it marked less than 1 per cent of either and only canopy height separated them. Backscatter and canopy height therefore marked the stands the beetle killed with no beetle data and no cruise in the spruce forest of Southcentral Alaska, but not in its open woodland. Of Alaska's 906 521 km² of forest, 29 per cent lay at or above the threshold, so the map shows managers which stands to protect first.

For Review Only

Introduction

The spruce beetle, *Dendroctonus rufipennis* (Kirby) [Coleoptera: Curculionidae: Scolytinae], is a bark beetle whose hosts are all the spruces of North America, 'with preference for large-diameter, mature trees' (Fettig et al. 2026, p. 1). Its brood takes one year or two to mature and summer warmth decides which, in that the larvae develop to adults within the first season when the bark is warm enough and otherwise pass the winter as larvae and emerge a year later (Hansen et al. 2001a). In Utah, Colorado, and on the Kenai Peninsula, the best predictor of a one-year brood was the number of hours above 17 °C in the 40 to 90 days after the adults flew (Hansen et al. 2001b), and a one-year cycle doubles the rate at which a population grows and spreads (Raffa et al. 2008, p. 511). Between outbreaks the beetle breeds in windthrown and felled spruce, which it prefers to standing trees (Schmid and Frye 1977, p. 11), and 'Once the availability of windthrown spruce is exhausted, mature adult spruce beetles often transition to attacking live spruces, which can lead to outbreaks' (Fettig et al. 2026, p. 1). A living spruce defends itself with resin, and its 'defenses are overcome only when a critical minimum number of beetles are recruited to the host tree' (Bentz et al. 2010, p. 603), so a small population takes only weakened trees, although 'Larger or thick-phloemed trees provide greater resources for larval feeding, and yield higher rates of beetle replacement' (Raffa et al. 2008, p. 512). Raffa et al. (2008, p. 501) therefore described bark beetle outbreaks as eruptions that 'occur when key thresholds are surpassed, prior constraints cease to exert influence, and positive feedbacks amplify across scales'. The large spruce is therefore the host the beetle gains most from and the one it reaches only in numbers, so it is taken first when an outbreak begins and the average size of the trees attacked falls as the outbreak runs (Massey and Wygant 1954, Schmid and Frye 1976).

In Alaska, the beetle's preferred hosts are Lutz spruce (*Picea* × *lutzii* Little) and white spruce (*P. glauca* (Moench) Voss) (Fettig et al. 2026), and its outbreaks have been more frequent and severe in Lutz spruce than in white or Sitka spruce (*P. sitchensis* (Bong.) Carrière) (Holsten and Werner 1990). They have recurred for as long as the record reaches. On the Kenai Peninsula, tree rings recorded regional outbreaks 'during the 1810s, 1850s, 1870–1880s, 1910s, and 1970–1980s', 'at a mean return interval of 52 years' (Berg et al. 2006, p. 219), and tree rings of 1601 to 2007 across south central and southwest Alaska gave

For Review Only

'evidence of regional-scale outbreaks dating from the mid-1700s', with the outbreaks since the 1990s, though within that range, showing 'greater spatiotemporal synchrony' than at any time in about 250 years (Sherriff et al. 2011, p. 1459). On the Kenai Peninsula, the beetle killed 29 per cent of the white spruce of the Resurrection Creek watershed between 1976 and 1980, taking the larger diameter classes first and smaller trees as the unattacked ran out (Werner and Holsten 1983), and across Alaska it likewise killed the larger spruce (Baker and Kemperman 1974, Holsten 1984), in the Copper River basin the more so the larger and the slower growing the tree (Doak 2004). On the same peninsula, the attacked trees had grown more slowly than the unattacked and a large diameter predicted attack or death only when coupled with slow growth (Hard et al. 1983, Hard 1985), and the Forest Service guide to those stands described the susceptible trees as 'often dominants or codominants in well-stocked stands' (Hard and Holsten 1985, p. 7), so the trees the beetle takes first are often the tallest of the stand.

Weather set their timing. On the Kenai Peninsula, the outbreaks of 1971 to 1996 'were positively associated with the 5-year backwards running average of summer temperature', and Berg et al. (2006, p. 219) suggested that warm temperature acted 'through a combination of increased overwinter survival, a doubling of the maturation rate from 2 years to 1 year, and regional drought-induced stress of mature host trees'. Across south central and southwest Alaska, 'El Niño years, combined with severe late-summer drought, appeared to contribute significantly to spruce beetle outbreaks', which the cool phase of the Pacific Decadal Oscillation, a pattern of Pacific Ocean temperatures that shifts every few decades, tended to precede (Sherriff et al. 2011, p. 1459). In Utah and Colorado, outbreak years followed Decembers and preceding Septembers about 2 °C warmer than other years and five years of below normal precipitation (Hebertson and Jenkins 2008, pp. 285–7), and in northwestern Colorado outbreaks were 'predicted best by interannual to multidecadal variability in drought, not by temperature alone' (Hart et al. 2014, p. 930). The warmth works only where there are large trees to take, in that the Kenai relationship 'decoupled after 1996, presumably because spruce beetles had killed most of the susceptible mature spruce in the region' (Berg et al. 2006, p. 219). From 1920 to 1989 about 847 000 ha of Alaska's spruce forest were infested, and the outbreak of 1990 to 2000 alone infested 1.19 million ha, 'approximately 40% more forest area than was infested the previous 70 years', which Werner et al.

For Review Only

(2006, p. 195) related 'to the relatively high densities of large-diameter spruce in an aging forest across the region and to a warming trend'. It peaked in 1996, when more than 500 000 ha of spruce stands were infested (Holsten and Hard 2001). The present outbreak has affected about 2 million acres of Southcentral Alaska, some 800 000 ha, since 2016, and had spread north to Cantwell and the Alaska Range by 2020 (Zwieback et al. 2024b).

Hazard ratings, scores of how likely a stand is to be attacked, applied that preference to the stand. The first was Research Note RM-309 of the Rocky Mountain Forest and Range Experiment Station, in which Schmid and Frye (1976, p. 1) rated Engelmann spruce (*Picea engelmannii* Parry ex Engelm.) stands 'on the basis of physiographic location, tree diameter, basal area, and percentage of spruce in the canopy', the basal area being the cross-sectional area of the stems per hectare, and set the high class at well-drained creek bottoms, an average diameter above 41 cm among the spruce over 25 cm, a basal area above 34 m² per hectare and more than 65 per cent spruce in the canopy. Its diameter classes were drawn from the White River outbreak of the 1940s in Colorado, in which the average diameter of the infested trees 'decreased from 21 to 17 to 15' inches as the large trees were used up (Schmid and Frye 1976, p. 2). On the Kenai Peninsula, a decision tree on basal area, spruce share, the share of the spruce basal area in trees over 25 cm in diameter, elevation, and aspect predicted high hazard correctly for at least 73 per cent of the stands (Reynolds and Holsten 1994a). The current Forest Service leaflet rates as most susceptible the stands in drainage bottoms whose spruce average more than 40 cm in diameter, whose basal area exceeds 34 m² per hectare and in which more than 60 per cent of the stand is live spruce (Fettig et al. 2026). A rating also falls as an outbreak proceeds, so a stand is rated before the beetle arrives. In central Utah, stands rated high to mostly medium hazard before the outbreak of 1986 to 1998 were rated medium to primarily low by 1998, after average spruce diameter, basal area, and overstory spruce had fallen (Dymerski et al. 2001). Every rating takes its diameter, basal area, and spruce share from a cruise, a ground inventory of the stand. The prevention a rating directs, thinning and partial cutting (Hard and Holsten 1985, Fettig et al. 2007, Hansen et al. 2010, Temperli et al. 2014), therefore needs the susceptible stands found first.

For Review Only

Remote sensing of the outbreak has mapped the damage after it appeared rather than the stands at risk. Landsat composites and dense time series mapped spruce beetle mortality in Colorado and its subtle change (Woodward et al. 2018, Ye et al. 2021), and the sensitivity of a Landsat series to mortality depended on its severity (Rodman et al. 2021). In the Susitna valley, about 90 per cent of the dead spruce mapped at crown scale on WorldView-2 imagery of 2021 stood in stands where dead crowns covered no more than 5 per cent of the ground (Zwieback et al. 2024b, p. 412), and Landsat reflectance change related clearly to that cover only above 10 per cent (Zwieback et al. 2024b). Radar was scarcely used for insect damage before Sentinel-1 (Hollaus and Vreugdenhil 2019), and in Bavaria it has since been fitted to the vitality of spruce attacked by bark beetles (Hechtl et al. 2025).

Radar backscatter and canopy height measure the structure of a stand. Backscatter, the part of the radar pulse that a stand reflects back to the satellite, rises with the dry mass of the trees until it saturates, at about 100 tons per hectare at L-band (Dobson et al. 1992), and Sentinel-1 operates at the shorter C-band, where 'the C-band backscattering coefficient shows much less sensitivity to total above-ground biomass' (Dobson et al. 1992, p. 412). Canopy height is now mapped worldwide at 10 m (Lang et al. 2023). A map of tall stands is therefore a map of the large dominant spruce that the hazard ratings describe, drawn without a cruise.

No study had scored radar backscatter or a canopy height map against the stands a spruce beetle outbreak killed, or mapped the stands susceptible to the beetle across a region. This study tested two hypotheses derived from that literature, and *Table 1* lists the measures entered, the attribute of the hazard ratings each stands for and the direction expected in killed stands. The first was that the Sentinel-1 backscatter of a stand before an outbreak and its canopy height would rank the stands the outbreak killed above those it spared, because both measure the large dominant spruce the beetle takes first, tested in the Susitna valley against the crown-scale map of dead spruce that Zwieback et al. (2024b) made from a WorldView-2 image of 2 August 2021. The second was that a statewide map of pre-outbreak backscatter would rank attacked above unattacked forest wherever the aerial detection survey, the yearly flights in which observers map forest damage from an aircraft, had recorded the outbreak, and where that test failed the map was to be narrowed to the regions in which it succeeded.

Commented [SM1]: Table 1 near here

For Review Only

Materials and methods

Study area

The study covered the forest of Alaska, 906 521 km² of 10 m cells with a canopy height of 5 m or more, which Sentinel-1 imaged from October 2014 (*Figure 1*). White and black spruce (*P. mariana* (Mill.) Britton et al.), Alaska birch (*Betula neoalaskana* Sarg.) and trembling aspen (*Populus tremuloides* Michx.) form the forest of the Interior and the Southcentral lowlands, with Lutz spruce on the Kenai Peninsula and Sitka spruce along the coast. The validation strip lay along the Parks Highway in the Susitna valley, between 62.1° N and 62.9° N, where Zwieback et al. (2024b) delineated dead spruce crowns. Within 20 km of the strip, the aerial detection survey mapped the most spruce beetle damage in 2018, 97 689 ha. The map was tested in four regions, the Kenai Peninsula Borough, which held most of the Alaska plot studies (Hard et al. 1983, Werner and Holsten 1983, Reynolds and Holsten 1994a), the Matanuska-Susitna Borough, which contains the strip, the Valdez-Cordova census area, which contains the Copper River basin of Doak (2004), and the Denali, Fairbanks North Star, and Southeast Fairbanks areas of the Interior, which the outbreak had reached at Cantwell by 2020 (Zwieback et al. 2024b).

Commented [SM2]: Figure 1 near here

Reference data

The reference data were the dead spruce crowns that Zwieback et al. (2024b) delineated by hand on a WorldView-2 image of 2 August 2021, 57 training areas of 512 m with 9 193 crowns and 121 test areas of 90 m with 3 147, and their severity map, the fraction of each 90 m cell covered by dead spruce, reported with a root mean square error of 0.02. The severity map has no declared licence and was used for validation only. The 2 crowns of the areas they delineated on a pre-outbreak image of 14 July 2013 were set aside. The delineators were ‘not able to directly identify a tree’s species, whether it was alive or dead, or the mortality agent when assembling the test data’ (Zwieback et al. 2024b, p. 418), so a crown is a dead spruce in an area of outbreak rather than a confirmed beetle kill. It records that the spruce stood dead on that date and not the year it died. Each area was divided into 10 m cells on the WGS 84 UTM zone 5N grid of the labels, 159 182 cells in all, and the fraction of each covered by dead crowns was computed. The aerial detection survey of the Forest Service gave the second label, its damage polygons recording survey year, host, and mortality class and its surveyed-area polygons

For Review Only

recording the ground the observers flew over. The Forest Inventory and Analysis plot and tree tables for Alaska (Bechtold and Patterson 2005) gave the ground check.

Satellite data

Sentinel-1 ground range detected scenes in interferometric wide swath mode were read in Google Earth Engine (Gorelick et al. 2017), where each had been corrected for thermal noise, calibrated, terrain corrected, and converted to decibels. Following the analysis-ready preparation of Mullissa et al. (2021), backscatter was normalized for slope with the volume scattering model of Vollrath et al. (2020), which also masks layover and radar shadow, on the Copernicus 30 m elevation model, because north of 60° N Mullissa et al. (2021) advised an alternative to the Shuttle Radar Topography Mission. Speckle was reduced with a 3 by 3 mean in linear power. The strip is imaged by the ascending orbit 65 and the descending orbit 131, which view the forest from opposite sides at different incidence angles, so the two were kept apart in the validation and pooled after a correction for incidence angle in the statewide map. Canopy height came from three sources. The ETH Zurich model of 2020 fitted the heights measured by the Global Ecosystem Dynamics Investigation lidar on the International Space Station to Sentinel-2 imagery at 10 m (Lang et al. 2023), and two surface models less the 10 m terrain model of the United States Geological Survey 3D Elevation Program gave the other two, the Copernicus GLO-30 surface from X-band radar of 2011 to 2015 and the ArcticDEM mosaic from optical stereo at 2 m (Porter et al. 2023). The yearly mosaics of the Phased Array L-band Synthetic Aperture Radar (PALSAR) on the Advanced Land Observing Satellite (ALOS) of the Japan Aerospace Exploration Agency gave L-band backscatter at 25 m in HH and HV, the horizontal pulse with its horizontal or vertical return (Shimada et al. 2014), and the LANDFIRE existing vegetation type gave the vegetation class of each cell (Rollins 2009). The conifer top cover of 2020 from the Arctic-Boreal Vulnerability Experiment (Macander et al. 2022) defined living forest.

Literature review

The literature behind the hypotheses was reviewed under a protocol frozen on 14 October 2026, with questions on the tree and stand attributes that predict attack, on the remote sensing of spruce beetle damage, and on whether a susceptibility map had been made before from radar or canopy height data

For Review Only

without a field cruise. Queries built from the title and keywords were run that day in the author's reference store, the Semantic Scholar bulk search, and OpenAlex and returned 337 records, 190 of them unique, one round of backward and forward citation chasing from five central records added 160, and 9 were added by hand. One reader screened all 359 on title and abstract and retained 97, every retained record behind an expected direction was read in full where its text could be reached, and the 21 whose text could not be reached are cited only for what their abstracts state.

Stand validation

The first hypothesis was tested on the stands of the label areas and then on the landscape. For every month from October 2014 to August 2021, the month of the label image, the median slope-corrected backscatter of each orbit was composited in Earth Engine and sampled at the centre of every 10 m cell of every label area. From June 2017, backscatter in VV and VH, the polarizations that send a vertical pulse and receive its vertical or horizontal return, was available on both orbits, and before then VV on both orbits with VH on the ascending orbit in scattered months from April 2015.

The test was made at the 90 m scale at which Zwieback et al. (2024b) reported severity. Each 512 m label area was divided into 90 m cells, and the 1 543 cells wholly inside their area took the mean dead crown fraction and the mean backscatter, in linear power, of their 81 cells of 10 m. A stand was dead at a dead crown fraction of 0.01 or more and living at 0 with conifer top cover of 5 per cent or more, so that the living class contained forest and not treeless ground. Because Zwieback et al. (2024b) related the change in Landsat shortwave infrared reflectance to severity, the change in VV between October 2014 to December 2016, before the survey recorded damage in any label area, and June 2019 to August 2021 was scored beside the level of each period. Each was scored by the area under the receiver operating characteristic curve, the chance that a killed stand drawn at random ranked above a living one, and the change was also tested with a Welch t-test, so that a difference that existed before the outbreak could be told from a change the outbreak made. To test whether the full monthly series added anything to the level, the temporal convolutional network of Pelletier et al. (2019), as implemented in sits (Simoes et al. 2021), was trained on the series of each stand. The 57 training areas were dealt to five spatial folds

For Review Only

of 15 km hexagons by blockCV (Valavi et al. 2018), so that no stand was scored by a network that had seen its area.

Landscape validation

Because the 1 543 stands were chosen by Zwieback et al. (2024b), the first hypothesis was then tested across their severity map, a strip 16.2 by 49 km. The mean slope-corrected VV of every scene of each orbit in the two periods, the three canopy heights, and the L-band mosaics of 2015 and 2016 were averaged into the 90 m cells of the map. Of the 84 038 cells with values, a cell was dead at a severity of 0.01 or more and living at 0 with conifer top cover of 5 per cent or more. Each source was scored as for the stands, and by Spearman's rank correlation with severity over the forest cells. Logistic regressions on the cells with every source present tested whether a second source improved the ranking given by the first. The level was scored against the survey polygons of 2017 to 2024 rasterized onto the same cells. The LANDFIRE class of each cell tested whether the height contrast was present within a cover class or was a contrast between host and non-host.

Statewide map

For the second hypothesis, every Sentinel-1 VV scene over Alaska from October 2014 to December 2016 was pooled into one map. Slope-corrected VV falls with incidence angle across a swath, so the fall was fitted as the slope of VV on angle over the forest cells of 150 scenes drawn at random, at 300 m, and the median over the 131 scenes that covered forest, -0.125 dB per degree, brought every scene to the mid-swath angle of 38 degrees. Forest was a 10 m cell with an ETH Zurich canopy height of 5 m or more. The product was computed at 90 m on the Alaska Albers grid, each scene read at the 90 m level of its image pyramid and averaged over scenes in linear power, with the scene count, the mean canopy height, and the forest share of the 10 m cells as further bands, and was exported as the Earth Engine asset named in the code. It was validated as made, read on the cells of the severity map and scored against the dead and living cells, the survey, and the ETH Zurich height. The threshold that maximized the index of Youden (1950), the sum of the two correct rates less one, -8.44 dB, defined a susceptible cell in everything that follows, the same rule gave 17.4 m on the height map, and the two correct rates and the overall accuracy, the share of all labelled cells classed correctly, were computed at that threshold.

For Review Only

Every source on the severity map, and the network on the crown labels, was then scored by the metrics of Murphy et al. (2026). Before any model was fitted, the separability of the dead from the living cells on each source was measured by the index M of Kaufman and Remer (1994), the difference of the two class means over the sum of their standard deviations, with M above 1 taken as good separability, and tested with the Wilcoxon rank-sum test. The area under the curve carried the 95 per cent confidence interval of DeLong et al. (1988), and at the Youden threshold of each source the confusion matrix gave Cohen's kappa (Cohen 1960), the commission and omission error of each class, and the overall accuracy with its exact binomial 95 per cent confidence interval. The threshold classifier and the network assumed no distribution of the backscatter, while the logistic fits were parametric binomial regressions, each scored by Nagelkerke's R^2 (Nagelkerke 1991) with the likelihood ratio test, the Hosmer-Lemeshow test over ten groups (Hosmer and Lemesbow 1980), the Brier score (Brier 1950), the root mean square error, the same error relative to the mean of the label, the mean absolute error, Moran's I of the residuals over the eight nearest cells (Moran 1950) and the Durbin-Watson statistic of the residuals in cell order (Durbin and Watson 1950). Overfitting was the root mean square error of the held-out folds over that of the training folds, rejected above 1.1 as in Murphy et al. (2026), and systematic bias the bias share of Theil's U (Theil 1966) on the held-out predictions, both over five spatial folds dealt by blockCV in blocks of 1 km, no smaller than the 0.81 km autocorrelation range of the sources on the strip. The network's ratio used the predictions of each fold's network on its own training stands.

Regional validation

The second hypothesis was tested against the aerial detection survey, the only label beyond the strip, in the four regions of *Figure 1*. In each region, a 90 m grid was laid on the Alaska Albers projection, and the survey's spruce beetle polygons on white spruce of 2016 to 2024 gave each cell its first year inside a polygon and the highest mortality class recorded over it. A cell was dead if it lay inside such a polygon and had conifer top cover of 5 per cent or more, and living if it had that cover, lay inside the ground flown in those years, had never lain inside a spruce beetle polygon of any host or year, and lay within 10 km of a white spruce polygon, so that the living class held forest the beetle could have reached and the surveyors had seen. The cells of the severity map were excluded. In each region, 5 000 dead and

For Review Only

5 000 living cells were drawn at random with a fixed seed, or every cell where fewer existed, and scored by the area under the curve and by the share of each class at or above the strip's threshold. Spearman's rank correlation between the year a dead cell was first recorded and its canopy height tested whether the beetle moved from taller to shorter stands, as Werner and Holsten (1983) recorded on the Kenai Peninsula and Massey and Wygant (1954) on the White River. In the Matanuska-Susitna Borough, the nine single sources of the strip, with the PALSAR mosaics of 2007 to 2010 that predate every damage polygon, were also read at 20 000 dead and 20 000 living cells drawn the same way, and scored again with the dead cells restricted to polygons of 20 per cent mortality or more, to measure the ceiling the survey label set on any source.

Ground plots

The second hypothesis was checked on the ground against the public Forest Inventory and Analysis plots of Alaska (Bechtold and Patterson 2005), comparing the dead with the live white spruce of each plot as the plot studies compared killed and surviving trees (Werner and Holsten 1983, Doak 2004). Every forested plot of the four regions was taken at its latest measurement, 129 plots with white spruce in the Matanuska-Susitna Borough (measured 2018 to 2020), 262 on the Kenai Peninsula (1999 to 2025), 136 in the Valdez-Cordova census area (2002 to 2025) and 317 in the Interior with the Yukon-Koyukuk area added (2014 to 2019). The Interior plots were measured before the survey recorded the outbreak's front along the Nenana River, 99.8 per cent of whose cells it first recorded in 2020 or later, so they stood for white spruce the beetle had not yet reached. The inventory records the diameter and height of every live or standing dead tree and a decay class for the dead, but no cause or year of death. On the plots holding both live and dead white spruce, the mean diameter and height of the dead trees were set against those of the live trees with a paired Wilcoxon signed-rank test. Across all plots, the height of the tallest tree was correlated with the share of white spruce basal area dead and ranked the plots with at least half of that basal area dead by the area under the curve. The radar and height sources were read at each plot's published coordinate, which the inventory shifts by up to about 1.6 km, for the 90 m cell holding it and as the mean of the cells within 800 m.

For Review Only

Results

Stand validation

At the 90 m stand scale, on 510 training stands labelled dead (369) or living forest (141), the mean VV level of October 2014 to December 2016, before the outbreak reached the areas, ranked the stands the outbreak would kill above the living stands with an area under the curve of 0.92 on the descending orbit and 0.89 on the ascending (*Table 2*). The dead stands were 1.9 dB brighter before the outbreak. The change from that level to June 2019 to August 2021 separated nothing, an area under the curve of 0.50 to 0.55, with the dead stands changing by -0.20 dB against -0.14 dB for the living [Welch t-test, t(169) = -1.35, p = 0.178]. Conifer cover alone gave 0.55, so the pre-outbreak brightness was not canopy density. The network trained on the full monthly series reached an F1 score of 0.92 and an area under the curve of 0.91 on held-out folds. Its root mean square error was 0.12 on its own training stands against 0.28 on the held-out folds, a ratio of 2.33 above the 1.1 line of Murphy et al. (2026), so the network was overfitted. The untrained mean of one band matched it at 0.92, so the level alone was carried forward.

Commented [SM3]: Table 2 near here

Landscape validation

Across the 84 038 cells of the severity map, 34 990 dead and 6 774 living forest, the pre-outbreak VV level ranked dead above living with an area under the curve of 0.88 on the ascending orbit and 0.80 on the descending. Its rank correlation with severity over every forest cell was 0.43 and 0.29 on the two orbits, respectively, and the mean level rose with every severity bin from -9.48 dB at zero to -7.76 dB at 0.1 or more on the descending orbit, while the change between the two periods gave 0.48 and 0.62 on the same orbits. The three canopy heights ranked the cells as well as the radar did, the ETH Zurich height at 0.89, with mean height of 24.7 m for the dead cells against 12.9 m for the living, the GLO-30 surface minus terrain at 0.84 and the ArcticDEM surface minus terrain at 0.84. L-band HV of 2015 and 2016 gave 0.86, and height and the C-band level were correlated over the forest cells [Spearman's rho 0.64]. On the 41 764 cells with every source, logistic fits gave an area under the curve of 0.887 for height alone, 0.884 for C-band alone, and 0.918 for all three together. Within every LANDFIRE class with at least 100 cells of each kind, height ranked dead above living, with areas under the curve from 0.67 to 0.95, and in Alaska Sub-boreal White Spruce Forest, the one white spruce class with enough cells, height

For Review Only

gave 0.86, with mean height of 21.0 m for the dead cells against 13.7 m for the living, while logistic fits gave 0.808 for the class alone and 0.904 for class and height together.

No single source reached a separability of M = 1 on the severity map, the index running from 0.54 for GLO-30 surface minus terrain to 0.90 for ETH canopy height 2020, though every source separated the two classes [Wilcoxon rank-sum, $p < 0.001$] (*Table 3*). At their Youden thresholds the sources classed 81 to 89 per cent of the labelled cells correctly, with Cohen's kappa of 0.42 to 0.61, and the commission error of the dead class stayed below 7 per cent while that of the smaller living class ran from 38 to 56 per cent. The logistic fits reached Nagelkerke's R^2 of 0.27 to 0.59, Brier scores of 0.064 to 0.105, and root mean square errors of 0.25 to 0.32 against a mean label of 0.84, 30.3 to 38.7 per cent of that mean, with mean absolute errors of 0.13 to 0.21, Theil's U bias shares of 0.000 to 0.000, and ratios of held-out to training error of 1.000 to 1.002, so none was overfitted, while the Hosmer-Lemeshow test rejected the calibration of every fit and the residuals were clustered in space [Moran's I 0.34 to 0.53, $p < 0.001$; Durbin-Watson 1.07 to 1.42] (*Table 4*).

Commented [SM4]: Table 3 near here

Commented [SM5]: Table 4 near here

Statewide map

The pooled product, read on the 41 764 labelled cells of the severity map from a median of 57 scenes per cell, ranked dead above living with an area under the curve of 0.86, with mean VV of -7.75 dB for the dead cells against -9.10 dB for the living. Its rank correlation with ETH Zurich height over the forest cells was 0.71. At the threshold of -8.44 dB, it called 91 per cent of the dead cells and 24 per cent of the living cells susceptible, an overall accuracy of 89 per cent. Against the survey polygons on the same cells it gave 0.67. Across Alaska (*Figure 2*), the map covered 906 521 km² of forest, 29 per cent of it at or above the threshold. The share ranged from 27 per cent in the Interior region to 62 per cent in the Kenai Peninsula region (*Table 5*).

Commented [SM6]: Figure 2 near here

Commented [SM7]: Table 5 near here

Regional validation

Against the aerial detection survey, the radar map ranked attacked above unattacked forest with an area under the curve of 0.70 on the Kenai Peninsula and 0.70 in the Matanuska-Susitna Borough, where the attacked stands were 0.8 dB brighter and 5.7 m taller than the unattacked. In the Copper River basin, where the survey had recorded damage only since 2022, it gave 0.44, and both classes were darker (-

For Review Only

10.1 against -10.0 dB) and shorter (13.6 against 10.4 m) than in the two southern regions. In the Interior, it gave 0.61. The 5 000 dead cells of that sample all lay between 63.3° N and 63.8° N along the Nenana River, and 98 per cent of them were first recorded in 2021 or later. The height map gave 0.67, 0.70, 0.76, and 0.58 in the same four regions, respectively (*Table 6*). In the Matanuska-Susitna Borough, the nine single sources scored from 0.50 to 0.72 against the same label, the highest being L-band HV of 2007 to 2010. Limiting the dead cells to polygons of 20 per cent mortality or more changed nothing, 0.51 to 0.70, so the survey label set the ceiling. The stands the survey first recorded in 2016 and 2017 were the tallest, 24.9 and 24.8 m on ETH Zurich height, and those first recorded in 2024 the shortest, 19.8 m, with Spearman's rho between year and height of -0.23 over cohorts of 645 to 4 417 cells.

Commented [SM8]: Table 6 near here

Ground plots

On the ground, the trees that died were the large ones. In the Matanuska-Susitna Borough, on the 77 plots holding both live and dead white spruce, the dead trees averaged 24.2 cm in diameter and 14.6 m in height against 15.7 cm and 10.4 m for the live trees [paired Wilcoxon signed-rank test, $V = 2\,811$, $p < 0.001$ for diameter and $V = 2\,779$, $p < 0.001$ for height]. Across the 129 plots, the share of white spruce basal area standing dead rose with the height of the tallest tree [Spearman's rho 0.60], which ranked the 75 plots with at least half their basal area dead with an area under the curve of 0.84. On the Kenai Peninsula, the dead trees on the 173 plots holding both were again the larger, 22.0 cm and 13.2 m against 17.8 cm and 11.2 m for the live trees [$V = 12\,213$, $p < 0.001$ and $V = 11\,673$, $p < 0.001$]. In the Copper River basin, the dead trees were 19.3 cm in diameter against 16.8 cm on 69 plots [$V = 1\,566$, $p = 0.032$], with no difference in height [$V = 1\,404.5$, $p = 0.240$]. On the Interior plots, the dead white spruce on 110 plots holding both were no thicker than the live ones, 17.3 against 17.5 cm [$V = 2\,557.5$, $p = 0.184$], and were shorter, 11.3 against 12.0 m [$V = 2\,079.5$, $p = 0.004$]. The ETH Zurich canopy height around a plot's published coordinate ranked the half-dead plots at 0.74 in the Matanuska-Susitna Borough, against 0.84 for the plot's own tallest tree.

Discussion

Over the Susitna valley label areas of Zwieback et al. (2024b), the Sentinel-1 VV level of October 2014 to December 2016 ranked the stands the outbreak would kill above those it left alive, while the change

For Review Only

through the outbreak in the same cells ranked nothing, so the first hypothesis was supported against the best ground truth this outbreak has, and what the level validated was a measurement of the stand before the outbreak, not a detection of mortality. The product's rank correlation with the ETH Zurich height over the forest was 0.71, so the radar level measured stand height and the map is a map of tall stands, and height, the C-band level, and L-band HV each ranked the cells in the direction the review expected.

Stand height

That the tall stands died is the oldest finding in the spruce beetle literature. On the Kenai Peninsula, 'Mortality was greatest in the larger diameter classes during the early part of the infestation' (Werner and Holsten 1983, p. 96), in the Copper River basin, 'Mortality increased with increasing DBH and decreasing BAI' (Doak 2004, p. 810), that is with larger diameter and slower growth, and the Forest Service synthesis reduced the plot studies to the rule that 'Spruce beetles preferentially attack and kill large diameter trees that have been growing slowly at breast height' (Hard and Holsten 1985, p. 7). The present study found the rule at the scale of the stand, from free radar and canopy height data with no beetle in the predictor, and the inventory plots confirmed it in the same decade, in that the dead white spruce stood 8.5 cm thicker and 4.2 m taller than the live trees beside them in the Matanuska-Susitna Borough, while on the Interior plots measured before the outbreak's front arrived the dead trees were no larger, so the size effect belongs to the beetle. The qualification of Hard et al. (1983, p. 678), that diameter mattered 'unless large diameter was coupled with slower than average radial growth in the last 5 years', is one the map does not include, because none of the measures here records growth, and height alone ranked the strip's stands at 0.89 without it. The order of attack was the same as on the White River (Schmid and Frye 1976, p. 2), in that the stands the survey recorded first in the Matanuska-Susitna Borough, in 2016 and 2017, were the tallest and each year's new stands were shorter through to 2024. The contrast was present inside each LANDFIRE class with enough cells, so it was a contrast between tall and short stands of the same cover and not between host and non-host.

Hazard ratings

The second hypothesis applied that finding to the hazard rating, which since Schmid and Frye (1976, p. 1) has scored stands 'on the basis of physiographic location, tree diameter, basal area, and percentage

For Review Only

of spruce in the canopy'. Those ratings take their diameter, basal area, and spruce share from a cruise, and the thresholds now in use (Fettig et al. 2026, p. 8) have never been applied to the forest of a state. The map gives the hazard part of such a rating at 90 m across 906 521 km² of forest from free data. Of that forest, 29 per cent lay at or above the threshold that separated the killed strip stands from the spared, from 27 per cent in the Interior to 62 per cent on the Kenai Peninsula (*Table 5*). It is hazard and not risk, in the distinction Reynolds and Holsten (1994b) drew when a panel of experts set stand hazard and windthrown trees as 'the two most important factors determining risk of a spruce beetle outbreak', because the map includes no windthrow, no beetle population, and no spruce share. A tall birch or aspen stand therefore scores as a tall spruce stand would, and a manager reads the map with the stand map in hand. What it directs is the prevention Hard and Holsten (1985, p. 2) prescribed, 'Mark all spruce trees 14 inches (36 cm) or larger in diameter for removal regardless of growth rate', before the beetle arrives.

Aerial surveys

Against the aerial detection survey, the map ranked attacked above unattacked forest at 0.70 on the Kenai Peninsula and 0.70 in the Matanuska-Susitna Borough. In the borough, every one of nine sources scored between 0.50 and 0.72 against the survey label, where the crown map had scored them at 0.88 to 0.89. The survey is drawn from an aircraft at a scale at which a polygon of 2 per cent mortality is mostly living forest, and restricting the dead cells to polygons of 20 per cent mortality or more changed nothing. The ceiling is therefore the label's and the regional numbers are a lower bound on the map's skill. In the Copper River basin, the radar level failed where the height map did not, 0.44 against 0.76, because the attacked stands there were 13.6 m tall against 22.1 m in the Matanuska-Susitna Borough and both classes averaged near -10 dB. The radar map is therefore validated for the Kenai Peninsula and the Susitna lowlands, and the height map is the product for the Copper River basin, the narrowing of extent the design allowed for. In the Interior, the map scored 0.61 at the front of the outbreak along the Nenana River, where 98 per cent of the recorded cells date from 2021 or later and the polygons contain stands the beetle has entered and not yet killed through.

For Review Only

Limitations

The crown labels and severity map are one image of one valley on one date, and the threshold was set on that valley, so a region whose stands are taller or shorter will have more or fewer cells above it than the beetle will take. The ETH Zurich height is a 2020 model whose estimates 'saturate around 25 to 30 m' (Lang et al. 2023, p. 1779), just above the mean 24.7 m of the strip's dead stands. The GLO-30 and ArcticDEM surfaces, the 2007 to 2010 PALSAR mosaic, and the 2014 to 2016 radar predate the outbreak and gave the same ranking. The pooling of orbits depends on one fitted fall of backscatter with incidence angle, -0.125 dB per degree, and on 49 to 72 scenes behind a forest cell in the four regions. Crown-scale detection in Alaska needs airborne hyperspectral and lidar data to reach '78% accuracy for all three health statuses' (Cessna et al. 2021), and Landsat detects the outbreak only above a severity of 0.1 (Zwieback et al. 2024b). The map offered here detects no outbreak at all, and claims only to say, before one arrives, where it will go.

Conclusion

Validated against the crown labels and severity map of Zwieback et al. (2024b), the pre-outbreak Sentinel-1 level and three independent canopy heights ranked the stands the spruce beetle killed in the Susitna valley above those it spared, the finding the plot studies have reported since 1954 at the scale of the tree, found here at the scale of the stand with no beetle in the predictor. Pooled across every scene over Alaska, the level gave a 90 m map of susceptible stands that ranked attacked above unattacked forest on the Kenai Peninsula and in the Susitna lowlands, where the inventory plots confirmed it on the ground, and failed in the open woodland of the Copper River basin, where the height map ranked them instead. The two maps give the hazard part of a stand rating with no cruise, and show forest managers which stands to protect first in the next outbreak.

For Review Only

Supplementary material

The following supplementary material is available at Forestry online. Figure S1 gives the area of spruce beetle damage mapped by the aerial detection survey in each survey year by quarter-degree band of latitude.

Funding

This work received no specific funding.

Conflict of interest

The author declares no conflict of interest.

Author contributions

The sole author designed the study, carried out the analysis and wrote the manuscript.

Data availability statement

Every input was public and was read by the code shown in this manuscript, except the severity map of Zwieback (2023), which carries no declared licence and was used for validation only. The crown labels of Zwieback et al. (2024b) came from Zenodo (Zwieback et al. 2024a) under the licence given in the code, the satellite, terrain, vegetation, and boundary layers were read in Google Earth Engine under the asset names given in the code, the aerial detection survey and the Forest Inventory and Analysis tables came from the United States Forest Service, and the conifer top cover from the Oak Ridge National Laboratory Distributed Active Archive Center. The 90 m statewide map is held as the Earth Engine asset named in the code. The sampled cells, the screening table of the literature review, the tables and figures are held with the manuscript in a public repository at *https://github.com/seamusrobertmurphy/deep-learn-alaska-spruce-beetle*.

For Review Only

References

Baker BH, Kemperman JA. Spruce beetle effects on a White spruce stand in Alaska. Journal of Forestry 1974;72:423–5. *https://doi.org/10.1093/jof/72.7.423*.

Bechtold WA, Patterson PL. The enhanced forest inventory and analysis program: National sampling design and estimation procedures. General Technical Report SRS-80. Asheville, NC: U.S. Department of Agriculture, Forest Service, Southern Research Station, 2005. *https://doi.org/10.2737/SRS-GTR-80*.

Bentz BJ, Régnière J, Fettig CJ et al. Climate change and bark beetles of the western United States and Canada: Direct and indirect effects. BioScience 2010;60:602–13. *https://doi.org/10.1525/bio.2010.60.8.6*.

Berg EE, David Henry J, Fastie CL et al. Spruce beetle outbreaks on the Kenai Peninsula, Alaska, and Kluane National Park and Reserve, Yukon Territory: Relationship to summer temperatures and regional differences in disturbance regimes. Forest Ecology and Management 2006;227:219–32. *https://doi.org/10.1016/j.foreco.2006.02.038*.

Brier GW. Verification of forecasts expressed in terms of probability. Monthly Weather Review 1950;78:1–3. *https://doi.org/10.1175/1520-0493(1950)078<0001:vofeit>2.0.co;2*.

Cessna J, Alonzo MG, Foster AC et al. Mapping boreal Forest spruce beetle health status at the individual crown scale using fused spectral and structural data. Forests 2021;12:1145. *https://doi.org/10.3390/f12091145*.

Cohen J. A coefficient of agreement for nominal scales. Educational and Psychological Measurement 1960;20:37–46. *https://doi.org/10.1177/001316446002000104*.

DeLong ER, DeLong DM, Clarke-Pearson DL. Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. Biometrics 1988;44:837. *https://doi.org/10.2307/2531595*.

Doak P. The impact of tree and stand characteristics on spruce beetle (Coleoptera: Scolytidae) induced mortality of white spruce in the Copper River Basin, Alaska. Canadian Journal of Forest Research 2004;34:810–6. *https://doi.org/10.1139/x03-256*.

Dobson MC, Ulaby FT, LeToan T et al. Dependence of radar backscatter on coniferous forest biomass. IEEE Transactions on Geoscience and Remote Sensing 1992;30:412–5. *https://doi.org/10.1109/36.134090*.

Durbin J, Watson GS. Testing for serial correlation in least squares regression. I. Biometrika 1950;37:409–28. *https://doi.org/10.1093/biomet/37.3-4.409*.

Dymerski AD, Anhold JA, Munson AS. Spruce beetle (Dendroctonus rufipennis) outbreak in Engelmann spruce (Picea engelmannii) in central Utah, 1986–1998. Western North American Naturalist 2001;61:19–24.

Fettig CJ, Audley JP, Malesky D et al. *Spruce beetle*. Forest Insect and Disease Leaflet 127. Washington, DC: USDA Forest Service, Forest Health Protection, 2026.

Fettig CJ, Klepzig KD, Billings RF et al. The effectiveness of vegetation management practices for prevention and control of bark beetle infestations in coniferous forests of the western and southern United States. Forest Ecology and Management 2007;238:24–53. *https://doi.org/10.1016/j.foreco.2006.10.011*.

Gorelick N, Hancher M, Dixon M et al. Google Earth engine: Planetary-scale geospatial analysis for everyone. Remote Sensing of Environment 2017;202:18–27. *https://doi.org/10.1016/j.rse.2017.06.031*.

For Review Only

Hansen EM, Bentz BJ, Turner DL. Temperature-based model for predicting univoltine brood proportions in spruce beetle (Coleoptera: Scolytidae). The Canadian Entomologist 2001b;133:827–41. *https://doi.org/10.4039/ent133827-6*.

Hansen EM, Bentz BJ, Turner DL. Physiological basis for flexible voltinism in the spruce beetle (Coleoptera: Scolytidae). The Canadian Entomologist 2001a;133:805–17. *https://doi.org/10.4039/ent133805-6*.

Hansen EM, Negron JF, Munson AS et al. A retrospective assessment of partial cutting to reduce spruce beetle-caused mortality in the Southern Rocky Mountains. Western Journal of Applied Forestry 2010;25:81–7. *https://doi.org/10.1093/wjaf/25.2.81*.

Hard JS. Spruce beetles attack slowly growing spruce. Forest Science 1985;31:839–50. *https://doi.org/10.1093/forestscience/31.4.839*.

Hard JS, Holsten EH. *Managing white and Lutz spruce stands in south-central Alaska for increased resistance to spruce beetle*. U.S. Department of Agriculture, Forest Service, Pacific Northwest Forest and Range Experiment Station, 1985. *https://doi.org/10.2737/pnw-gtr-188*.

Hard JS, Werner RA, Holsten EH. Susceptibility of white spruce to attack by spruce beetles during the early years of an outbreak in Alaska. Canadian Journal of Forest Research 1983;13:678–84. *https://doi.org/10.1139/x83-097*.

Hart SJ, Veblen TT, Eisenhart KS et al. Drought induces spruce beetle (Dendroctonus rufipennis) outbreaks across northwestern Colorado. Ecology 2014;95:930–9. *https://doi.org/10.1890/13-0230.1*.

Hebertson EG, Jenkins MJ. Climate factors associated with historic spruce beetle (Coleoptera: Curculionidae) outbreaks in Utah and Colorado. Environmental Entomology 2008;37:281–92. *https://doi.org/10.1093/ee/37.2.281*.

Hechtl C, Hauser S, Schmitt A et al. Kennaugh elements allow early detection of bark beetle infestation in temperate forests using Sentinel-1 data. Forests 2025;16:1272. *https://doi.org/10.3390/f16081272*.

Hollaus M, Vreugdenhil M. Radar satellite imagery for detecting bark beetle outbreaks in forests. Current Forestry Reports 2019;5:240–50. *https://doi.org/10.1007/s40725-019-00098-z*.

Holsten EH. Factors of susceptibility in spruce beetle attack on white spruce in Alaska. Journal of the Entomological Society of British Columbia 1984;81:39–45.

Holsten EH, Hard JS. *Dispersal flight and attack of the spruce beetle, Dendroctonus rufipennis, in south-central Alaska*. U.S. Department of Agriculture, Forest Service, Pacific Northwest Research Station, 2001. *https://doi.org/10.2737/pnw-rp-536*.

Holsten EH, Werner RA. Comparison of white, Sitka, and Lutz spruce as hosts of the spruce beetle in Alaska. Canadian Journal of Forest Research 1990;20:292–7. *https://doi.org/10.1139/x90-043*.

Hosmer DW, Lemesbow S. Goodness of fit tests for the multiple logistic regression model. Communications in Statistics - Theory and Methods 1980;9:1043–69. *https://doi.org/10.1080/03610928008827941*.

Kaufman YJ, Remer LA. Detection of forests using mid-IR reflectance: An application for aerosol studies. IEEE Transactions on Geoscience and Remote Sensing 1994;32:672–83. *https://doi.org/10.1109/36.297984*.

Lang N, Jetz W, Schindler K et al. A high-resolution canopy height model of the Earth. Nature Ecology & Evolution 2023;7:1778–89. *https://doi.org/10.1038/s41559-023-02206-6*.

For Review Only

Macander MJ, Nelson PR, Nawrocki TW et al. Time-series maps reveal widespread change in plant functional type cover across Arctic and boreal Alaska and Yukon. Environmental Research Letters 2022;17:054042. *https://doi.org/10.1088/1748-9326/ac6965*.

Massey CL, Wygant ND. *Biology and control of the Engelmann spruce beetle in Colorado*. Circular 944. Washington, DC: US Department of Agriculture, 1954.

Moran PAP. Notes on continuous stochastic phenomena. Biometrika 1950;37:17–23. *https://doi.org/10.1093/biomet/37.1-2.17*.

Mullissa A, Vollrath A, Odongo-Braun C et al. Sentinel-1 SAR backscatter analysis ready data preparation in Google Earth engine. Remote Sensing 2021;13:1954. *https://doi.org/10.3390/rs13101954*.

Murphy S, Leslie A, Wilson J et al. Spatial patterns of conifer regeneration following Mountain pine beetle outbreak (Dendroctonus ponderosae) and mixed-severity wildfire: A point process model of disturbance interactions in the southern Selkirk Mountains, British Columbia. Forest Ecology and Management 2026;618:123985. *https://doi.org/10.1016/j.foreco.2026.123985*.

Nagelkerke NJD. A note on a general definition of the coefficient of determination. Biometrika 1991;78:691–2. *https://doi.org/10.1093/biomet/78.3.691*.

Pelletier C, Webb G, Petitjean F. Temporal convolutional neural network for the classification of satellite image time series. Remote Sensing 2019;11:523. *https://doi.org/10.3390/rs11050523*.

Porter C, Howat I, Noh M-J et al. *ArcticDEM - mosaics, version 4.1*. Harvard Dataverse, 2023. *https://doi.org/10.7910/DVN/3VDC4W*.

Raffa KF, Aukema BH, Bentz BJ et al. Cross-scale drivers of natural disturbances prone to anthropogenic amplification: The dynamics of bark beetle eruptions. BioScience 2008;58:501–17. *https://doi.org/10.1641/b580607*.

Reynolds KM, Holsten EH. Relative importance of risk factors for spruce beetle outbreaks. Canadian Journal of Forest Research 1994b;24:2089–95. *https://doi.org/10.1139/x94-268*.

Reynolds KM, Holsten EH. Classification of spruce beetle hazard in Lutz spruce (Picea × lutzii) stands on the Kenai Peninsula, Alaska. Canadian Journal of Forest Research 1994a;24:1015–21. *https://doi.org/10.1139/x94-132*.

Rodman KC, Andrus RA, Veblen TT et al. Disturbance detection in Landsat time series is influenced by tree mortality agent and severity, not by prior disturbance. Remote Sensing of Environment 2021;254:112244. *https://doi.org/10.1016/j.rse.2020.112244*.

Rollins MG. LANDFIRE: A nationally consistent vegetation, wildland fire, and fuel assessment. International Journal of Wildland Fire 2009;18:235–49. *https://doi.org/10.1071/WF08088*.

Santoro M, Cartus O. Research pathways of Forest above-ground biomass estimation based on SAR backscatter and interferometric SAR observations. Remote Sensing 2018;10:608. *https://doi.org/10.3390/rs10040608*.

Schmid JM, Frye RH. *Stand ratings for spruce beetles*. Research Note RM-309. Fort Collins, Colorado: USDA Forest Service, Rocky Mountain Forest and Range Experiment Station, 1976.

Schmid JM, Frye RH. *Spruce beetle in the Rockies*. General Technical Report RM-49. Fort Collins, Colorado: USDA Forest Service, Rocky Mountain Forest and Range Experiment Station, 1977.

Sherriff RL, Berg EE, Miller AE. Climate variability and spruce beetle (Dendroctonus rufipennis) outbreaks in south-central and southwest Alaska. Ecology 2011;92:1459–70. *https://doi.org/10.1890/10-1118.1*.

For Review Only

Shimada M, Itoh T, Motooka T et al. New global forest/non-forest maps from ALOS PALSAR data (2007–2010). Remote Sensing of Environment 2014;155:13–31. *https://doi.org/10.1016/j.rse.2014.04.014*.

Simoes R, Camara G, Queiroz G et al. Satellite image time series analysis for big Earth observation data. Remote Sensing 2021;13:2428. *https://doi.org/10.3390/rs13132428*.

Temperli C, Hart SJ, Veblen TT et al. Are density reduction treatments effective at managing for resistance or resilience to spruce beetle disturbance in the southern Rocky Mountains? Forest Ecology and Management 2014;334:53–63. *https://doi.org/10.1016/j.foreco.2014.08.028*.

Theil H. *Applied economic forecasting*. Amsterdam: North-Holland, 1966.

Valavi R, Elith J, Lahoz-Monfort JJ et al. blockCV: An R package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. Methods in Ecology and Evolution 2018;10:225–32. *https://doi.org/10.1111/2041-210x.13107*.

Veblen TT, Hadley KS, Reid MS et al. The response of subalpine forests to spruce beetle outbreak in Colorado. Ecology 1991;72:213–31. *https://doi.org/10.2307/1938916*.

Vollrath A, Mullissa A, Reiche J. Angular-based radiometric slope correction for Sentinel-1 on Google Earth engine. Remote Sensing 2020;12:1867. *https://doi.org/10.3390/rs12111867*.

Werner RA, Holsten EH. Mortality of white spruce during a spruce beetle outbreak on the Kenai Peninsula in Alaska. Canadian Journal of Forest Research 1983;13:96–101. *https://doi.org/10.1139/x83-014*.

Werner RA, Holsten EH, Matsuoka SM et al. Spruce beetles and forest ecosystems in south-central Alaska: A review of 30 years of research. Forest Ecology and Management 2006;227:195–206. *https://doi.org/10.1016/j.foreco.2006.02.050*.

Woodward B, Evangelista P, Vorster A. Mapping progression and severity of a Southern Colorado spruce beetle outbreak using calibrated image composites. Forests 2018;9:336. *https://doi.org/10.3390/f9060336*.

Ye S, Rogan J, Zhu Z et al. Detecting subtle change from dense Landsat time series: Case studies of mountain pine beetle and spruce beetle disturbance. Remote Sensing of Environment 2021;263:112560. *https://doi.org/10.1016/j.rse.2021.112560*.

Youden WJ. Index for rating diagnostic tests. Cancer 1950;3:32–5. *https://doi.org/10.1002/1097-0142(1950)3:1<32::aid-cncr2820030106>3.0.co;2-3*.

Zwieback S, White J, Morris M. *[dataset] Spruce beetle training and validation data*. Zenodo, 2024a. *https://doi.org/10.5281/zenodo.10569990*.

Zwieback S, Young-Robertson J, Robertson M et al. Low-severity spruce beetle infestation mapped from high-resolution satellite imagery with a convolutional network. ISPRS Journal of Photogrammetry and Remote Sensing 2024b;212:412–21. *https://doi.org/10.1016/j.isprsjprs.2024.05.013*.

For Review Only

Table 1. The measures the study entered, the attribute of the hazard ratings each stands for, the direction expected in the stands an outbreak killed, and the literature the expectation rests on.

Measure	Indicator	Expected	Basis
Canopy height before the outbreak, from three sources	Large dominant spruce	Taller	The beetle takes large, slowly growing dominants and codominants first (Massey and Wygant 1954, Werner and Holsten 1983, Hard and Holsten 1985, Doak 2004), and an outbreak left stands with smaller average and maximum heights (Veblen et al. 1991)
Sentinel-1 C-band VV level before the outbreak	Stand structure and biomass	Higher	Backscatter rises with forest biomass until it saturates, C-band being much less sensitive than L-band (Dobson et al. 1992, Santoro and Cartus 2018)
ALOS PALSAR L-band HV level	Stand structure and biomass	Higher	Backscatter rises with biomass to a saturation near 100 tons per hectare at L-band (Dobson et al. 1992)
Change in VV through the outbreak	Mortality	A difference of either sign, if the radar recorded the mortality	Radar was proposed for bark beetle damage, although detection before the crowns fade has had limited success (Hollaus and Vreugdenhil 2019, Hechtl et al. 2025)
Diameter and height of dead and live white spruce on inventory plots	Tree size	Dead trees larger and taller where the outbreak has run	Killed trees were larger and grew more slowly than survivors in the same stands (Werner and Holsten 1983, Hard 1985, Doak 2004)
Year the survey first recorded a stand	Order of attack	Taller stands recorded earlier	The largest trees died first and smaller ones as the outbreak proceeded (Massey and Wygant 1954, Schmid and Frye 1976, Werner and Holsten 1983)

For Review Only

Table 2. Area under the receiver operating characteristic curve for ranking the stands the outbreak killed above the living stands, by source and by label. The crown labels are the 90 m stand cells of the Zwieback et al. (2024) training areas, the severity map their 90 m map of 2021, and the survey the aerial detection survey polygons of 2016 to 2024 across the rest of the Matanuska-Susitna Borough. The statewide product is the pooled, angle-corrected VV level of every scene; the two orbits are the single-orbit levels of the same months.

Source	Crown labels	Severity map	Survey, Matanuska-Susitna
Sentinel-1 VV, statewide product		0.86	
Sentinel-1 VV, ascending orbit 65	0.89	0.88	0.63
Sentinel-1 VV, descending orbit 131	0.92	0.80	0.61
ETH canopy height 2020		0.89	0.70
GLO-30 surface minus terrain		0.84	0.64
ArcticDEM surface minus terrain		0.84	0.50
PALSAR L-band HV		0.86	0.72
Conifer top cover	0.55		

For Review Only

*Table 3. Separability and classification accuracy of each source at its own Youden threshold against the dead and living cells of the severity map, and of the temporal network at a probability of 0.5 on the held-out folds of the crown labels. M is the separability index of Kaufman and Remer (1994), the difference of the class means over the sum of their standard deviations, starred by the Wilcoxon rank-sum test; AUC is the area under the receiver operating characteristic curve with the 95 per cent confidence interval of DeLong et al. (1988); the threshold is in the source's own unit, decibels for radar, metres for height and probability for the network; κ is Cohen's kappa; commission and omission errors are given for each class; OA is the overall accuracy with its exact binomial 95 per cent confidence interval. * $p \le 0.05$, ** $p \le 0.01$, *** $p \le 0.001$, **** $p \le 0.0001$.*

Source	Dead	Living	M	AUC (95% CI)	Threshold	κ	Commission dead	Commission living	Omission dead	Omission living	OA (95% CI)
Sentinel-1 VV, statewide product	34,990	6,774	0.61****	0.86 (0.86, 0.87)	-8.44	0.61	0.05	0.38	0.09	0.24	0.89 (0.88, 0.89)
Sentinel-1 VV, ascending orbit 65	34,990	6,774	0.81****	0.88 (0.87, 0.88)	-9.04	0.59	0.05	0.42	0.11	0.23	0.87 (0.87, 0.88)
Sentinel-1 VV, descending orbit 131	34,990	6,774	0.56****	0.80 (0.80, 0.81)	-8.84	0.42	0.07	0.56	0.17	0.30	0.81 (0.80, 0.81)
ETH canopy height 2020	34,990	6,774	0.90****	0.89 (0.88, 0.89)	17.45	0.53	0.04	0.49	0.15	0.19	0.84 (0.84, 0.85)
GLO-30 surface minus terrain	34,990	6,774	0.54****	0.84 (0.83, 0.84)	0.42	0.50	0.05	0.52	0.16	0.22	0.83 (0.83, 0.83)
ArcticDEM surface minus terrain	34,990	6,774	0.62****	0.84 (0.83, 0.85)	12.20	0.47	0.06	0.52	0.15	0.29	0.83 (0.82, 0.83)
PALSAR L-band HV	34,990	6,774	0.84****	0.86 (0.86, 0.87)	-14.08	0.56	0.05	0.44	0.11	0.25	0.86 (0.86, 0.87)
PALSAR L-band HH	34,990	6,774	0.65****	0.82 (0.81, 0.83)	-8.68	0.53	0.06	0.45	0.11	0.32	0.86 (0.86, 0.86)
Temp network, crown labels, held-out folds	369	141		0.91 (0.88, 0.94)	0.50	0.70	0.10	0.16	0.05	0.28	0.88 (0.85, 0.91)

*Table 4. Fit and overfitting diagnostics of the logistic fits of the dead and living cells of the severity map on each source and combination, and of the temporal network on the crown labels. R^2 is Nagelkerke's, starred by the likelihood ratio test against the null fit; HL is the Hosmer-Lemeshow statistic over ten groups, starred by its test; Brier is the mean squared error of the fitted probability against the label, RMSE its root, rRMSE the root relative to the mean of the label in per cent and MAE the mean absolute error; the RMSE ratio is the root mean square error of the held-out folds over that of the training folds across five spatial folds, 1.1 being the line above which a fit is overfitted (Murphy et al. 2026); U bias is the bias share of Theil's U on the held-out predictions; I is Moran's I of the residuals over the eight nearest cells, starred by its test; DW is the Durbin-Watson statistic of the residuals in cell order. * $p \le 0.05$, ** $p \le 0.01$, *** $p \le 0.001$, **** $p \le 0.0001$.*

Model	n	R^2	HL	Brier	RMSE	rRMSE (%)	MAE	RMSE ratio	U bias	I	DW
Sentinel-1 VV, statewide product	41,764	0.423****	2074****	0.075	0.274	32.7	0.162	1.00	0.000	0.432****	1.26
Sentinel-1 VV, ascending orbit 65	41,764	0.487****	641****	0.074	0.273	32.6	0.154	1.00	0.000	0.444****	1.24
Sentinel-1 VV, descending orbit 131	41,764	0.267****	667****	0.105	0.324	38.7	0.213	1.00	0.000	0.476****	1.25
ETH canopy height 2020	41,764	0.456****	212****	0.086	0.293	34.9	0.173	1.00	0.000	0.451****	1.21
GLO-30 surface minus terrain	41,764	0.274****	3198****	0.103	0.321	38.4	0.214	1.00	0.000	0.529****	1.07
PALSAR L-band HV	41,764	0.464****	369****	0.080	0.283	33.7	0.160	1.00	0.000	0.356****	1.38
Two canopy heights	41,764	0.465****	681****	0.082	0.286	34.2	0.169	1.00	0.000	0.455****	1.20
Two C-band orbits	41,764	0.517****	621****	0.069	0.263	31.4	0.144	1.00	0.000	0.427****	1.25
L-band HV and HH	41,764	0.474****	265****	0.080	0.282	33.7	0.159	1.00	0.000	0.336****	1.42
C-band and heights	41,764	0.559****	133****	0.068	0.261	31.1	0.139	1.00	0.000	0.425****	1.25
L-band and heights	41,764	0.539****	199****	0.072	0.269	32.1	0.146	1.00	0.000	0.349****	1.39
All six sources	41,764	0.594****	44****	0.064	0.253	30.3	0.130	1.00	0.000	0.365****	1.35
Temp network, crown labels, held-out folds	510			0.094	0.306	42.3	0.147	2.33	0.007	0.248****	1.41

634

Table 5. Forest and susceptible forest in the four test regions and in Alaska, from the 1 km map. Forest is the area of 10 m cells with canopy height of 5 m or more; susceptible forest is the forest whose pre-outbreak VV level reached the strip's threshold, and tall forest the forest whose canopy height reached the strip's height threshold. Scenes is the median number of Sentinel-1 scenes behind a forest cell.

Region	Forest (km²)	Susceptible (km²)	Susceptible (%)	Tall (km²)	Scenes
Kenai Peninsula	25 856	16 139	62	6 366	49
Matanuska-Susitna	47 011	22 617	48	9 976	57
Copper River	51 186	25 754	50	7 439	63
Interior	81 395	21 885	27	4 174	72
Alaska	906 521	258 717	29	106 336	53

For Review Only

635

Table 6. The statewide product against the aerial detection survey in four regions. Dead cells lay inside a spruce beetle polygon on white spruce of 2016 to 2024; living cells were forest the surveyors flew over, never inside a spruce beetle polygon of any year, within 10 km of one. AUC is the area under the curve for ranking dead above living; the two shares are the cells of each class the strip's VV threshold called susceptible; rho is Spearman's correlation between the year a dead cell was first recorded and its canopy height.

Region	Dead	Living	AUC, VV	AUC, height	VV dead (dB)	VV living (dB)	Susceptible dead (%)	Susceptible living (%)	rho year, height
Kenai Peninsula	5 000	5 000	0.70	0.67	-8.1	-8.7	77	44	0.09
Matanuska-Susitna	5 000	5 000	0.70	0.70	-7.9	-8.7	80	48	-0.23
Copper River	1 719	5 000	0.44	0.76	-10.1	-10.0	1	0	0.23
Interior	5 000	2 800	0.61	0.58	-8.6	-8.9	40	27	-0.02

For Review Only

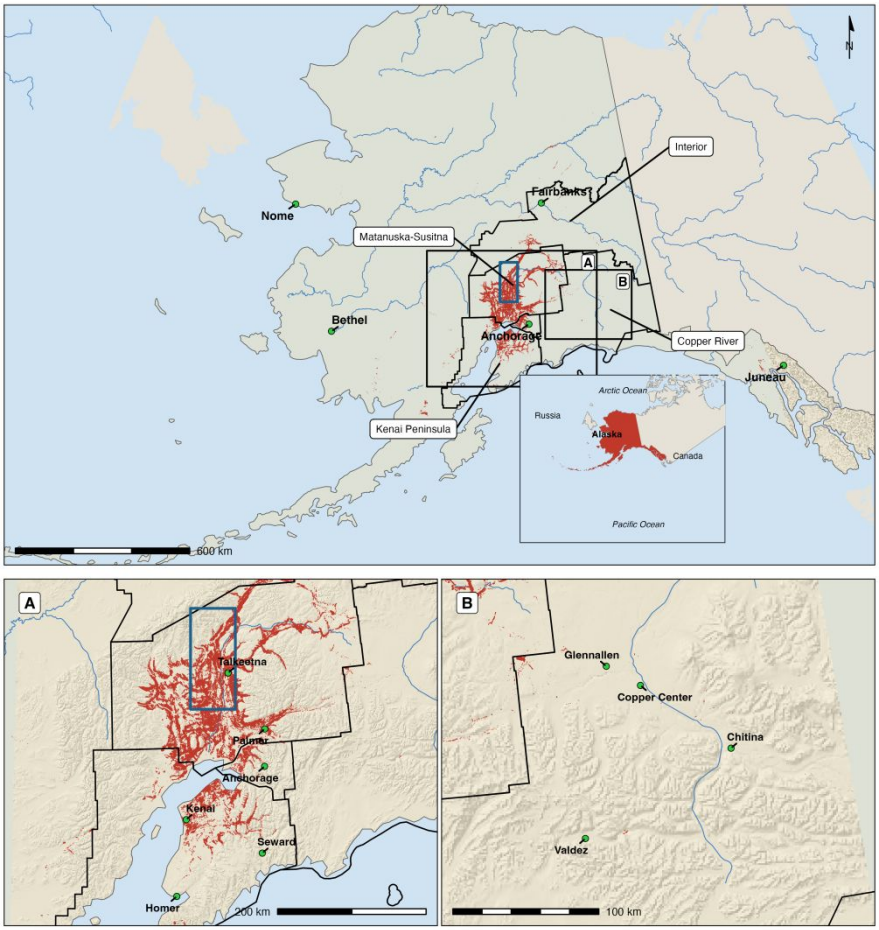


Figure 1. The study area. Main map, Alaska in the Alaska Albers projection (EPSG:3338) over a shaded relief, with the spruce beetle damage mapped by the aerial detection survey from 2016 to 2024 in red, the four regions in which the statewide map was tested outlined in black, the validation strip of the Susitna valley in which Zwieback et al. (2024) mapped dead spruce crowns on a WorldView-2 image of 2 August 2021 framed in blue, and the two detail panels boxed. A, Cook Inlet, the Kenai Peninsula and the Susitna valley. B, the Copper River basin. The inset places Alaska in North America.

636 Alt text: Three maps. The main map shows Alaska over shaded relief with the spruce beetle damage of
637 the 2016 to 2024 aerial detection survey in red, the four test regions outlined in black and labelled, the
638 Susitna valley validation strip framed in blue, major rivers in blue, the towns of Anchorage, Fairbanks,
639 Juneau, Nome and Bethel, two boxes marked A and B, a scale bar, a north arrow and an inset locating
640 Alaska between Russia and Canada. Panel A enlarges Cook Inlet, the Kenai Peninsula and the Susitna

641 valley with the towns of Anchorage, Kenai, Homer, Seward, Palmer and Talkeetna. Panel B enlarges
642 the Copper River basin with Glennallen and Valdez.

For Review Only

643

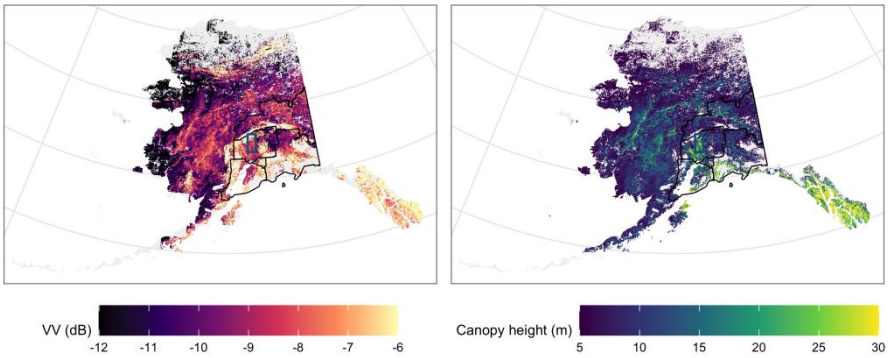


Figure 2. The statewide susceptibility map at 1 km, the mean of the 90 m product over the forest cells of each 1 km cell. Left, the slope-corrected Sentinel-1 VV level of October 2014 to December 2016 brought to a 38 degree incidence angle; right, the ETH canopy height of 2020. Cells with less than a fifth of their area in forest are blank. Black outlines are the four test regions and the blue frame the validation strip.

644 Alt text: Two maps of Alaska at 1 km. The left map shades the forest cells by their pre-outbreak Sentinel-

645 1 VV backscatter and the right map by their ETH canopy height of 2020, with the four test regions

646 outlined in black and the validation strip framed in blue.

For Review Only

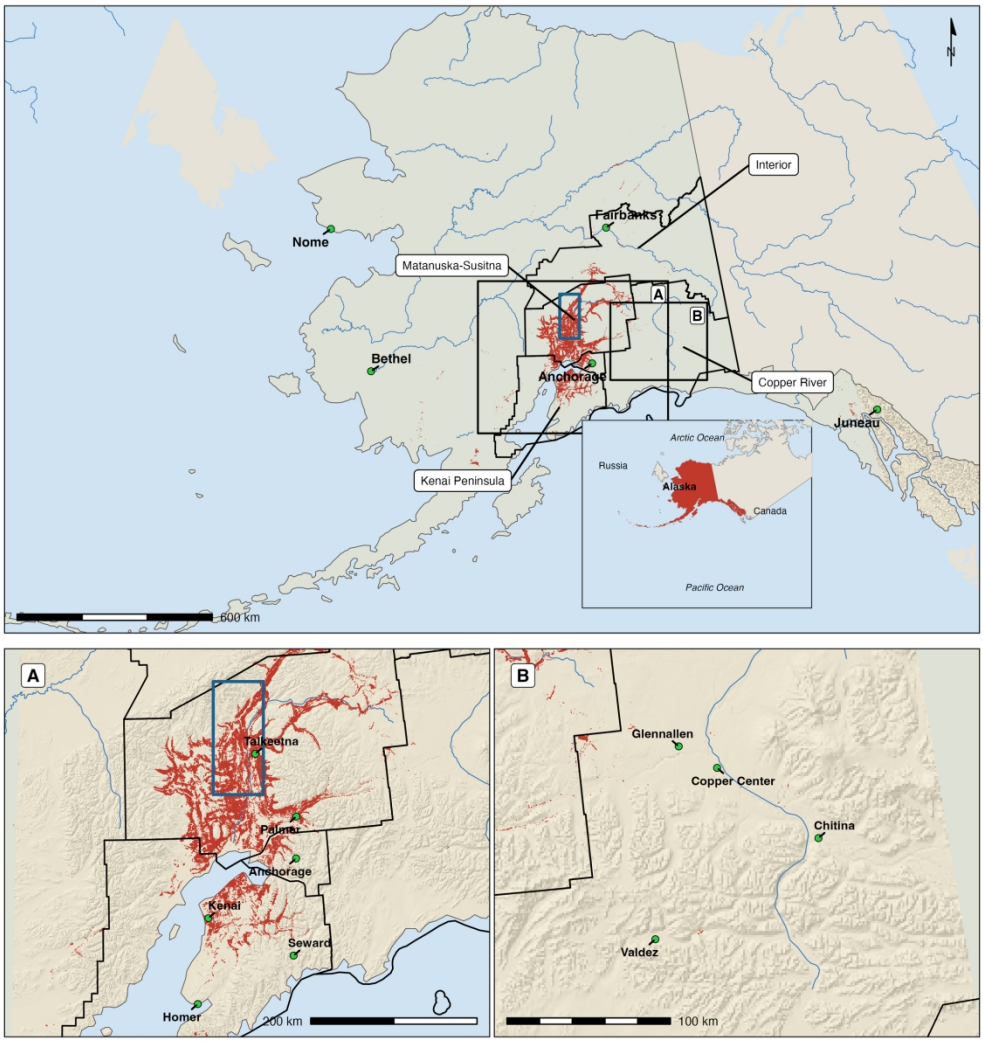


Figure 1. The study area. Main map, Alaska in the Alaska Albers projection (EPSG:3338) over a shaded relief, with the spruce beetle damage mapped by the aerial detection survey from 2016 to 2024 in red, the four regions in which the statewide map was tested outlined in black, the validation strip of the Susitna valley in which Zwieback et al. (2024) mapped dead spruce crowns on a WorldView-2 image of 2 August 2021 framed in blue, and the two detail panels boxed. A, Cook Inlet, the Kenai Peninsula and the Susitna valley. B, the Copper River basin. The inset places Alaska in North America.

190x203mm (300 x 300 DPI)

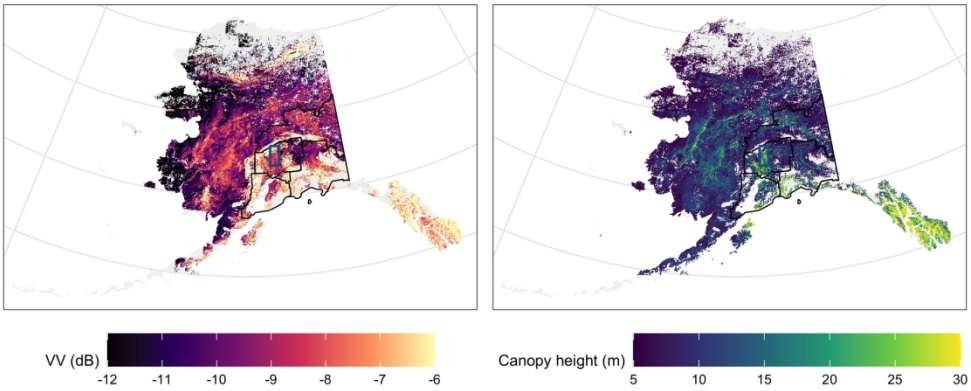


Figure 2. The statewide susceptibility map at 1 km, the mean of the 90 m product over the forest cells of each 1 km cell. Left, the slope-corrected Sentinel-1 VV level of October 2014 to December 2016 brought to a 38 degree incidence angle; right, the ETH canopy height of 2020. Cells with less than a fifth of their area in forest are blank. Black outlines are the four test regions and the blue frame the validation strip.

182x99mm (300 x 300 DPI)

1

For Review Only

Table 1. The measures the study entered, the attribute of the hazard ratings each stands for, the direction expected in the stands an outbreak killed, and the literature the expectation rests on.

Measure	Stands for	Expected in killed stands	Basis
Canopy height before the outbreak, from three sources	Large dominant spruce	Taller	The beetle takes large, slowly growing dominants and codominants first (Massey and Wygant 1954, Werner and Holsten 1983, Hard and Holsten 1985, Doak 2004), and an outbreak left stands with smaller average and maximum heights (Veblen et al. 1991)
Sentinel-1 C-band VV level before the outbreak	Stand structure and biomass	Higher	Backscatter rises with forest biomass until it saturates, C-band being much less sensitive than L-band (Dobson et al. 1992, Santoro and Cartus 2018)
ALOS PALSAR L-band HV level	Stand structure and biomass	Higher	Backscatter rises with biomass to a saturation near 100 tons per hectare at L-band (Dobson et al. 1992)

For Review Only

Change in VV through the outbreak	Mortality	A difference of either sign, if the radar recorded the mortality	Radar was proposed for bark beetle damage, although detection before the crowns fade has had limited success (Hollaus and Vreugdenhil 2019, Hechtl et al. 2025)
Diameter and height of dead and live white spruce on inventory plots	Tree size	Dead trees larger and taller where the outbreak has run	Killed trees were larger and grew more slowly than survivors in the same stands (Werner and Holsten 1983, Hard 1985, Doak 2004)
Year the survey first recorded a stand	Order of attack	Taller stands recorded earlier	The largest trees died first and smaller ones as the outbreak proceeded (Massey and Wygant 1954, Schmid and Frye 1976, Werner and Holsten 1983)

For Review Only

2

For Review Only

Table 2. Area under the receiver operating characteristic curve for ranking the stands the outbreak killed above the living stands, by source and by label. The crown labels are the 90 m stand cells of the Zwieback et al. (2024) training areas, the severity map their 90 m map of 2021, and the survey the aerial detection survey polygons of 2016 to 2024 across the rest of the Matanuska-Susitna Borough. The statewide product is the pooled, angle-corrected VV level of every scene; the two orbits are the single-orbit levels of the same months.

Source	Crown labels	Severity map	Survey, Matanuska-Susitna
Sentinel-1 VV, statewide product		0.86	
Sentinel-1 VV, ascending orbit 65	0.89	0.88	0.63
Sentinel-1 VV, descending orbit 131	0.92	0.80	0.61
ETH canopy height 2020		0.89	0.70
GLO-30 surface minus terrain		0.84	0.64
ArcticDEM surface minus terrain		0.84	0.50
PALSAR L-band HV		0.86	0.72
Conifer top cover	0.55		

For Review Only

3

For Review Only

*Table 3. Separability and classification accuracy of each source at its own Youden threshold against the dead and living cells of the severity map, and of the temporal network at a probability of 0.5 on the held-out folds of the crown labels. M is the separability index of Kaufman and Remer (1994), the difference of the class means over the sum of their standard deviations, starred by the Wilcoxon rank-sum test; AUC is the area under the receiver operating characteristic curve with the 95 per cent confidence interval of DeLong et al. (1988); the threshold is in the source's own unit, decibels for radar, metres for height and probability for the network; κ is Cohen's kappa; commission and omission errors are given for each class; OA is the overall accuracy with its exact binomial 95 per cent confidence interval. * $p \le 0.05$, ** $p \le 0.01$, *** $p \le 0.001$, **** $p \le 0.0001$.*

Source	Dead	Living	M	AUC (95% CI)	Threshold	κ	Commission dead	Commission living	Omission dead	Omission living	OA (95% CI)
Sentinel-1 VV, statewide product	34,990	6,774	0.61****	0.86 (0.86, 0.87)	-8.44	0.61	0.05	0.38	0.09	0.24	0.89 (0.88, 0.89)
Sentinel-1 VV, ascending orbit 65	34,990	6,774	0.81****	0.88 (0.87, 0.88)	-9.04	0.59	0.05	0.42	0.11	0.23	0.87 (0.87, 0.88)
Sentinel-1 VV, descending orbit 131	34,990	6,774	0.56****	0.80 (0.80, 0.81)	-8.84	0.42	0.07	0.56	0.17	0.30	0.81 (0.80, 0.81)
ETH canopy height 2020	34,990	6,774	0.90****	0.89 (0.88, 0.89)	17.45	0.53	0.04	0.49	0.15	0.19	0.84 (0.84, 0.85)
GLO-30 surface minus terrain	34,990	6,774	0.54****	0.84 (0.83, 0.84)	0.42	0.50	0.05	0.52	0.16	0.22	0.83 (0.83, 0.83)
ArcticDEM surface minus terrain	34,990	6,774	0.62****	0.84 (0.83, 0.85)	12.20	0.47	0.06	0.52	0.15	0.29	0.83 (0.82, 0.83)

PALSAR L-band HV	34,990	6,774	0.84****	0.86 (0.86, 0.87)	-14.08	0.56	0.05	0.44	0.11	0.25	0.86 (0.86, 0.87)
PALSAR L-band HH	34,990	6,774	0.65****	0.82 (0.81, 0.83)	-8.68	0.53	0.06	0.45	0.11	0.32	0.86 (0.86, 0.86)
Temp network, crown labels, held-out folds	369	141		0.91 (0.88, 0.94)	0.50	0.70	0.10	0.16	0.05	0.28	0.88 (0.85, 0.91)

For Review Only

4

For Review Only

*Table 4. Fit and overfitting diagnostics of the logistic fits of the dead and living cells of the severity map on each source and combination, and of the temporal network on the crown labels. R^2 is Nagelkerke's, starred by the likelihood ratio test against the null fit; HL is the Hosmer-Lemeshow statistic over ten groups, starred by its test; Brier is the mean squared error of the fitted probability against the label, RMSE its root, rRMSE the root relative to the mean of the label in per cent and MAE the mean absolute error; the RMSE ratio is the root mean square error of the held-out folds over that of the training folds across five spatial folds, 1.1 being the line above which a fit is overfitted (Murphy et al. 2026); U bias is the bias share of Theil's U on the held-out predictions; I is Moran's I of the residuals over the eight nearest cells, starred by its test; DW is the Durbin-Watson statistic of the residuals in cell order. * $p \le 0.05$, ** $p \le 0.01$, *** $p \le 0.001$, **** $p \le 0.0001$.*

Model	n	R^2	HL	Brier	RMSE	rRMSE (%)	MAE	RMSE ratio	U bias	I	DW
Sentinel-1 VV, statewide product	41,764	0.423****	2074****	0.075	0.274	32.7	0.162	1.00	0.000	0.432****	1.26
Sentinel-1 VV, ascending orbit 65	41,764	0.487****	641****	0.074	0.273	32.6	0.154	1.00	0.000	0.444****	1.24
Sentinel-1 VV, descending orbit 131	41,764	0.267****	667****	0.105	0.324	38.7	0.213	1.00	0.000	0.476****	1.25
ETH canopy height 2020	41,764	0.456****	212****	0.086	0.293	34.9	0.173	1.00	0.000	0.451****	1.21
GLO-30 surface minus terrain	41,764	0.274****	3198****	0.103	0.321	38.4	0.214	1.00	0.000	0.529****	1.07
PALSAR L-band HV	41,764	0.464****	369****	0.080	0.283	33.7	0.160	1.00	0.000	0.356****	1.38

Two canopy heights	41,764	0.465****	681****	0.082	0.286	34.2	0.169	1.00	0.000	0.455****	1.20
Two C-band orbits	41,764	0.517****	621****	0.069	0.263	31.4	0.144	1.00	0.000	0.427****	1.25
L-band HV and HH	41,764	0.474****	265****	0.080	0.282	33.7	0.159	1.00	0.000	0.336****	1.42
C-band and heights	41,764	0.559****	133****	0.068	0.261	31.1	0.139	1.00	0.000	0.425****	1.25
L-band and heights	41,764	0.539****	199****	0.072	0.269	32.1	0.146	1.00	0.000	0.349****	1.39
All six sources	41,764	0.594****	44****	0.064	0.253	30.3	0.130	1.00	0.000	0.365****	1.35
Temp network, crown labels, held-out folds	510			0.094	0.306	42.3	0.147	2.33	0.007	0.248****	1.41

5

For Review Only

Table 5. Forest and susceptible forest in the four test regions and in Alaska, from the 1 km map. Forest is the area of 10 m cells with canopy height of 5 m or more; susceptible forest is the forest whose pre-outbreak VV level reached the strip's threshold, and tall forest the forest whose canopy height reached the strip's height threshold. Scenes is the median number of Sentinel-1 scenes behind a forest cell.

Region	Forest (km²)	Susceptible (km²)	Susceptible (%)	Tall (km²)	Scenes
Kenai Peninsula	25 856	16 139	62	6 366	49
Matanuska-Susitna	47 011	22 617	48	9 976	57
Copper River	51 186	25 754	50	7 439	63
Interior	81 395	21 885	27	4 174	72
Alaska	906 521	258 717	29	106 336	53

For Review Only

6

For Review Only

Table 6. The statewide product against the aerial detection survey in four regions. Dead cells lay inside a spruce beetle polygon on white spruce of 2016 to 2024; living cells were forest the surveyors flew over, never inside a spruce beetle polygon of any year, within 10 km of one. AUC is the area under the curve for ranking dead above living; the two shares are the cells of each class the strip's VV threshold called susceptible; rho is Spearman's correlation between the year a dead cell was first recorded and its canopy height.

Region	Dead	Living	AUC, VV	AUC, height	VV dead (dB)	VV living (dB)	Susceptible dead (%)	Susceptible living (%)	rho year, height
Kenai Peninsula	5 000	5 000	0.70	0.67	-8.1	-8.7	77	44	0.09
Matanuska-Susitna	5 000	5 000	0.70	0.70	-7.9	-8.7	80	48	-0.23
Copper River	1 719	5 000	0.44	0.76	-10.1	-10.0	1	0	0.23
Interior	5 000	2 800	0.61	0.58	-8.6	-8.9	40	27	-0.02

7

Review Only